

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 25 minutes. Questions follow the six parts of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. Electronegativity values (Pauling): H 2.1, C 2.5, N 3.0, O 3.5, F 4.0, Cl 3.0, Na 0.9. A periodic table and the answer key are at the back.

Part 1-2 · Covalent bonds, formulas and names

Q1 IC2-LO1

In a neutral molecule, oxygen usually forms how many covalent bonds?

- A. 1
- B. 2
- C. 3
- D. 4

Q2 IC2-LO2

Which compound contains **only** covalent bonds?

- A. Na_2O
- B. $\text{Ba}(\text{OH})_2$
- C. N_2O_4
- D. NH_4Cl

Q3 IC2-LO2

Write the formula of **dinitrogen tetroxide**.

Q4 IC2-L02

Name the compound **PCl₅**.

Q5 IC2-L02

Name the compound **SO₃**.

Part 3 · Lewis structures

Q6 IC2-L03

How many valence electrons are used in the Lewis structure of the ammonium ion, **NH₄⁺**?

Q7 IC2-L03

Which molecule contains a **triple** bond?

- A. O₂
- B. CO₂
- C. N₂
- D. H₂O

Q8 IC2-L03

In the correct Lewis structure of CO₂, the carbon atom has:

- A. two single bonds and two lone pairs
- B. two double bonds and no lone pairs
- C. one triple bond and one single bond, no lone pairs
- D. four single bonds

Part 4 • Bond characteristics

Q9 IC2-LO4

Use the electronegativity difference to classify each bond (nonpolar < 0.4 · polar $0.4\text{--}1.8$ · ionic > 1.8).

Options: nonpolar covalent · polar covalent · ionic

H-H

O-H

Na-Cl

C-O

Q10 IC2-LO4

Rank these carbon-carbon bonds by length.

Rank 1 (shortest) to 3 (longest): C-C · C=C · C $\equiv$ C

Part 5 • Characteristics of molecules

Q11 IC2-LO5

Calculate the molecular mass of sulfur dioxide, SO₂ (S = 32.07, O = 16.00).

Q12 IC2-LO5

Match each molecule to its shape.

Options: linear · trigonal planar · tetrahedral · trigonal pyramidal · bent

CH₄

NH₃

H₂O

BF₃

CO₂

Q13 IC2-LO5

Which molecule is **nonpolar** even though its bonds are polar?

- A. NH_3
- B. H_2O
- C. CCl_4
- D. CH_3Cl

Part 6 • Intermolecular forces

Q14 IC2-LO6

The strongest intermolecular force between HF molecules is:

- A. dispersion forces
- B. dipole-dipole forces
- C. hydrogen bonding
- D. ionic bonding

Q15 IC2-LO6

Rank by boiling point.

Rank 1 (lowest) to 3 (highest): $\text{CH}_4 \cdot \text{HCl} \cdot \text{H}_2\text{O}$

Pharmacy scenario • Formalin

Scenario

Your hospital laboratory keeps tissue specimens in **formalin**, a water solution of formaldehyde, CH_2O (Ball 4.3). Carbon is the central atom, bonded to two H atoms and one O atom. Use this to answer the next four questions.

Q16 IC2-LO3

How many valence electrons are in the Lewis structure of CH_2O ?

Q17 IC2-LO5

What is the shape of formaldehyde around the carbon atom?

- A. linear
- B. trigonal planar
- C. tetrahedral
- D. bent

Q18 IC2-LO5

Calculate the molecular mass of CH_2O (C = 12.01, H = 1.01, O = 16.00).

Q19 IC2-LO6

Explain why formaldehyde dissolves well in water. Use the words *polar* and *hydrogen bond*.

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.85 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

After class: check your understanding

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About 25 minutes. Questions follow the six parts of the lecture and end with a pharmacy scenario.

Answer key

Part 1-2 · Covalent bonds, formulas and names

Q1 IC2-LO1

In a neutral molecule, oxygen usually forms how many covalent bonds?

- A. 1
- B. 2**
- C. 3
- D. 4

O has 6 valence electrons, needs 2 more: 2 bonds and 2 lone pairs (as in H_2O).

Q2 IC2-LO2

Which compound contains **only** covalent bonds?

- A. Na_2O
- B. $\text{Ba}(\text{OH})_2$
- C. N_2O_4**
- D. NH_4Cl

N_2O_4 has only nonmetals. $\text{Ba}(\text{OH})_2$ and NH_4Cl contain ions (both types); Na_2O is ionic.

Q3 IC2-LO2

Write the formula of **dinitrogen tetroxide**.

N_2O_4

di- = 2 N, tetr(a)- = 4 O.

Q4 IC2-LO2

Name the compound **PCl₅**.

phosphorus pentachloride

No mono- on the first element; penta- = 5; chlorine → chloride.

Q5 IC2-LO2

Name the compound **SO₃**.

sulfur trioxide

tri- = 3 oxygen atoms; oxygen → oxide.

Part 3 · Lewis structures

Q6 IC2-LO3

How many valence electrons are used in the Lewis structure of the ammonium ion, NH₄⁺?

8 electrons

N 5 + 4 × H 1 = 9, minus 1 for the + charge = 8.

Q7 IC2-LO3

Which molecule contains a **triple** bond?

A. O₂

B. CO₂

C. N₂

D. H₂O

N₂ has 10 valence electrons: :N≡N:. O₂ and CO₂ have double bonds.

Q8 IC2-LO3

In the correct Lewis structure of CO₂, the carbon atom has:

- A. two single bonds and two lone pairs
- B. two double bonds and no lone pairs**
- C. one triple bond and one single bond, no lone pairs
- D. four single bonds

O=C=O: 16 electrons, every atom has an octet, C has no lone pairs.

Part 4 • Bond characteristics

Q9 IC2-LO4

Use the electronegativity difference to classify each bond (nonpolar < 0.4 • polar 0.4–1.8 • ionic > 1.8).

H-H → nonpolar covalent; O-H → polar covalent; Na-Cl → ionic; C-O → polar covalent

Differences: 0 • 1.4 • 2.1 • 1.0.

Q10 IC2-LO4

Rank these carbon-carbon bonds by length.

C≡C < C=C < C-C (shortest → longest)

More shared pairs pull the nuclei closer: C≡C 120 pm, C=C 134 pm, C-C 154 pm.

Part 5 • Characteristics of molecules

Q11 IC2-LO5

Calculate the molecular mass of sulfur dioxide, SO₂ (S = 32.07, O = 16.00).

64.07 amu

32.07 + 2 × 16.00 = 64.07 amu.

Q12 IC2-LO5

Match each molecule to its shape.

CH₄ → tetrahedral; NH₃ → trigonal pyramidal; H₂O → bent; BF₃ → trigonal planar; CO₂ → linear

Count bonding groups and lone pairs on the central atom.

Q13 IC2-LO5

Which molecule is **nonpolar** even though its bonds are polar?

- A. NH₃
- B. H₂O
- C. CCl₄**
- D. CH₃Cl

CCl₄ is tetrahedral with four identical C-Cl bonds, so the dipoles cancel.

Part 6 • Intermolecular forces

Q14 IC2-LO6

The strongest intermolecular force between HF molecules is:

- A. dispersion forces
- B. dipole-dipole forces
- C. hydrogen bonding**
- D. ionic bonding

H is bonded directly to F, so HF molecules form hydrogen bonds.

Q15 IC2-LO6

Rank by boiling point.

CH₄ < HCl < H₂O (lowest → highest)

CH₄ dispersion only (−161 °C) < HCl dipole-dipole (−85 °C) < H₂O hydrogen bonding (100 °C).

Pharmacy scenario · Formalin

Q16 IC2-LO3

How many valence electrons are in the Lewis structure of CH₂O?

12 electrons

C 4 + 2 × H 1 + O 6 = 12. Lewis structure: C=O double bond, two C-H bonds, two lone pairs on O.

Q17 IC2-LO5

What is the shape of formaldehyde around the carbon atom?

A. linear

B. trigonal planar

C. tetrahedral

D. bent

3 bonding groups (the double bond counts as one), no lone pairs on C → trigonal planar, about 120°.

Q18 IC2-LO5

Calculate the molecular mass of CH₂O (C = 12.01, H = 1.01, O = 16.00).

30.03 amu

12.01 + 2 × 1.01 + 16.00 = 30.03 amu.

Q19 IC2-LO6

Explain why formaldehyde dissolves well in water. Use the words *polar* and *hydrogen bond*.

The C=O bond is polar (EN difference 3.5 – 2.5 = 1.0), and the trigonal planar shape does not cancel this dipole, so **formaldehyde is a polar molecule**. Water molecules form **hydrogen bonds** to the lone pairs on formaldehyde's oxygen, and the two polar molecules also attract by dipole-dipole forces. Polar dissolves polar ("like dissolves like"). Note: formaldehyde cannot hydrogen-bond to itself, because its H atoms are bonded to C, not to N, O or F.

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